SUMMARY · MK 2 FINANCIAL MANAGEMENT & STRATEGY · WEEK 3

Cash Flow & Financial Planning (Cash Budgets & Pro Forma)

After reading and analyzing the past, this week looks forward: measuring cash flow (depreciation, OCF, free cash flow), building a cash budget for short-term needs, and preparing pro forma statements. Based on Zutter & Smart ch. 4, applied in the Indofood 5-year forecast assignment.

Material

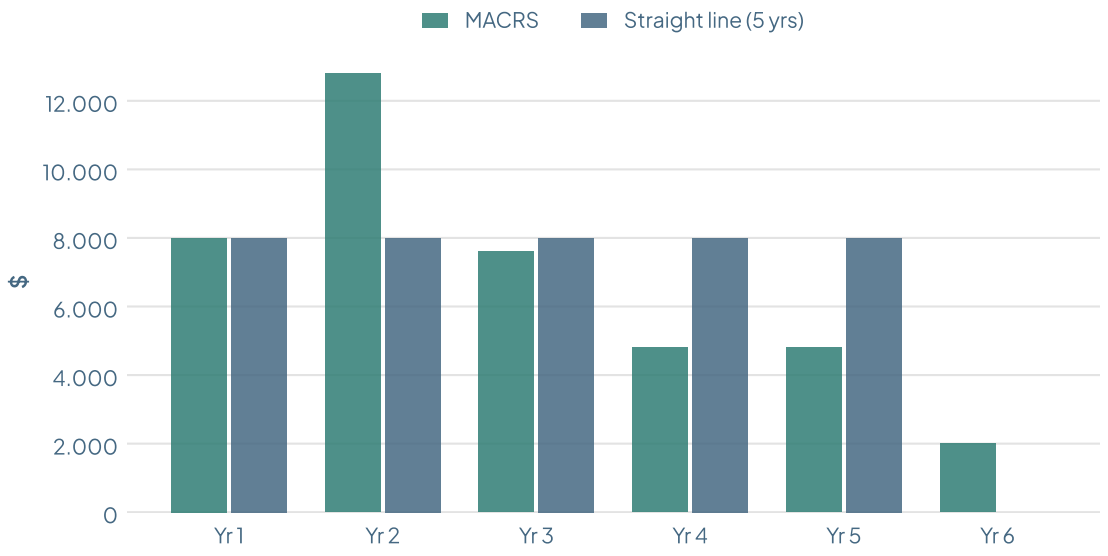
1 Depreciation: an expense that uses no cash

Depreciation spreads a fixed asset's cost over its life — a *noncash charge*. Depreciable value = cost + installation (Baker: $38k + $2k = $40k). US tax uses MACRS (5-year class: 20/32/19/12/12/5%). Indonesia's rules differ, but faster depreciation = lower early tax = more early cash.

2 Why faster depreciation means more cash early Deep dive

Depreciation creates a tax shield = depreciation × tax rate; MACRS front-loads it, which is worth more in present value.

Baker $40,000 machine: MACRS 5-year vs straight line




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Year 2: $12,800 × 21% = $2,688
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3 Measuring cash flow: sources vs uses, OCF, FCF

Sources	Uses
Asset decrease; liability increase; net profit; depreciation; stock sale	Asset increase; liability decrease; loss; dividends; buyback

Baker 2019: CFO 837, CFI −300, CFF +20 → cash +557. `OCF = EBIT(1-T) + Dep = 392.3` `FCF = OCF - NFAI - NCAI = 392.3 - 300 - 57 = 35.3` Positive but below $70 interest — not enough return for investors.

4 Cash budget: forecasting shortfalls and surpluses

A **cash budget** plans monthly receipts and disbursements, driven by the sales forecast (external or internal). Receipts − disbursements = net cash flow; + beginning cash = ending cash; vs minimum balance → required financing or excess cash.

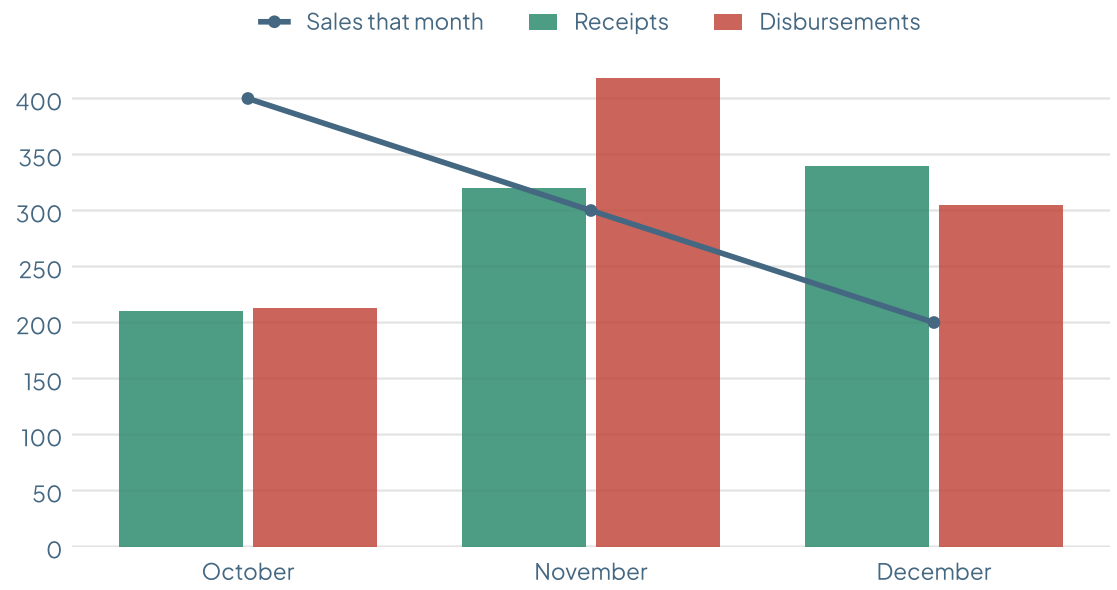
Coulson: Oct receipts 210/disbursements 213 → excess 22; Nov 320/418 → borrow 76; Dec 340/305 → repay to 41. Highest sales month (Oct) isn't the tightest cash month (Nov) — profit ≠ cash. Arrange credit lines in advance.

Scenario analysis: worst-case peak borrowing $190 vs most-likely $76. Monitor intra-month cash too. Same logic for personal budgets.

5 Coulson's cash budget charted: profit ≠ cash

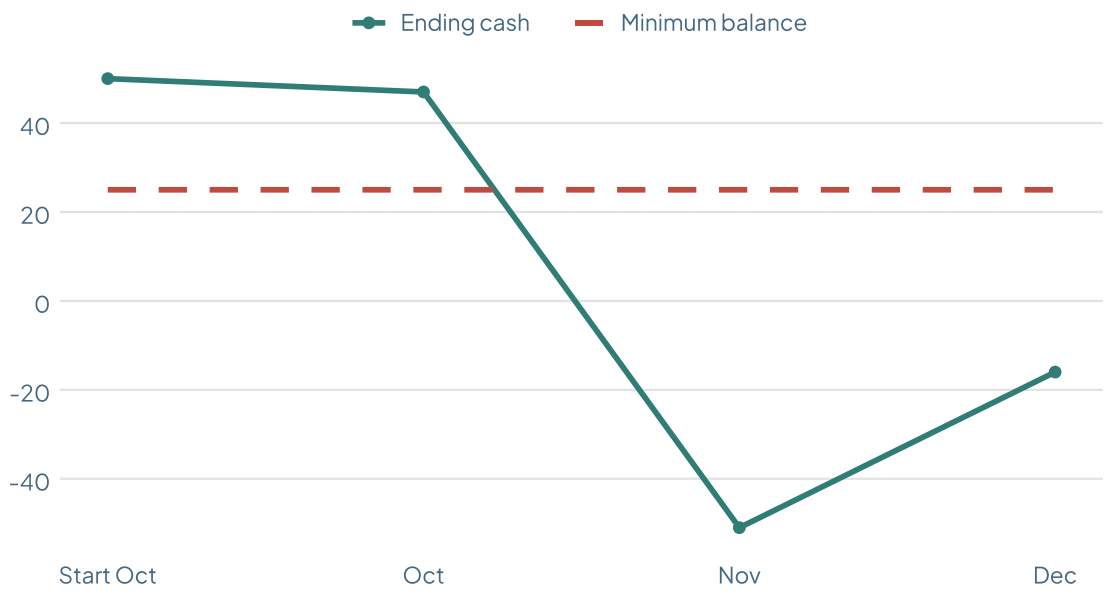
Deep dive

Coulson: cash receipts vs disbursements ($000)



Sales peak in October but cash is tightest in November (lagged collections, $130 machine).

Ending cash vs minimum balance



Plan for the worst case

Pessimistic peak borrowing: $190.

6 Pro forma statements: projected income and balance sheet

Fair Traders: sales $100k → $135k forecast.

Percent-of-sales income statement gives EBT $12,150 but assumes all costs variable. Splitting fixed/variable costs gives **$28,250** — operating leverage (see Week 5).

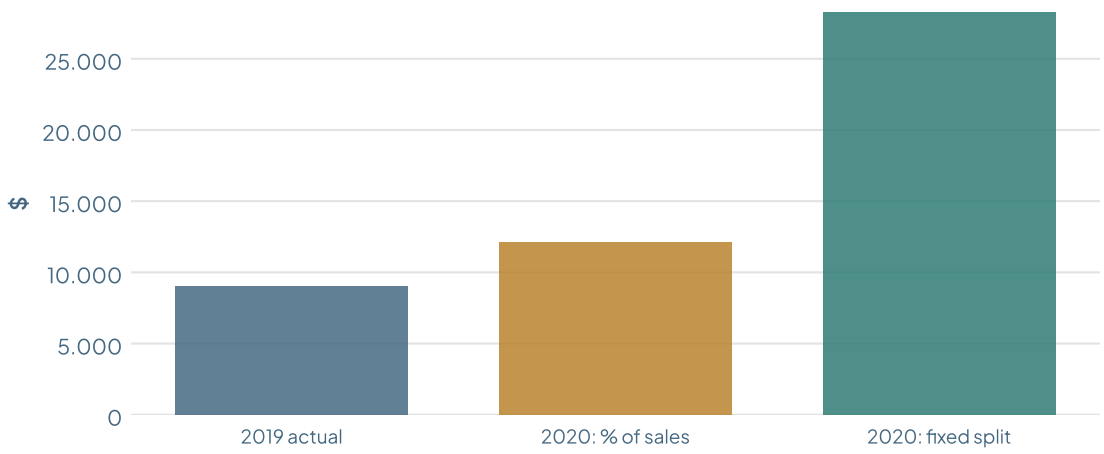
Judgmental balance sheet: estimate each line (cash $6k, receivables 45 days = $16,875, inventory $16k, net fixed assets $63k, payables 73 days = $8,100, taxes payable $455, retained earnings $29,327) → assets $105,875 vs funded $97,582 → **external financing required $8,293** (the plug).

Weaknesses: assumes the past repeats and forces some values; still valuable for projected ratios and adjusting plans.

7 Percent-of-sales vs fixed/variable split: a big difference

Deep dive

Fair Traders: projected 2020 EBT



Percent-of-sales gives $12,150 EBT; splitting fixed and variable costs gives $28,250 — operating leverage.

For the Indofood task

Separate truly variable costs (wheat, CPO, packaging) from fixed ones.

8 Why is Baker's CFO $837 but OCF $392.3? Two 'operating cash flow' numbers

In the slides, Baker Corporation has two numbers that are both called operating cash flow. This often confuses students, so let us untangle it.

Cash flow statement — operating section ($000)	Amount	Logic
Net profit after taxes	237	Starting point
Depreciation	+100	Non-cash charge, added back
Decrease in receivables	+100	Asset down = cash in

Decrease in inventories	+300	Asset down = cash in
Increase in accounts payable	+200	Liability up = cash in
Decrease in accruals	−100	Liability down = cash out
Cash provided by operating activities (CFO)	**837**	

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OCF = NOPAT + Depreciation = EBIT × (1 - T) + Depreciation = 370 × 0.79 + 100 = 392.3
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The difference:

- **CFO (837)** includes working-capital changes and is already after interest. This year Baker "released" a lot of cash from inventory and receivables, so CFO looks very large — but that cannot be repeated every year.
- **OCF (392.3)**, the finance version, measures cash from *core operations* before interest and before working-capital changes. Working capital is counted separately in NCAI, so FCF = OCF – NFAI – NCAI = 392.3 – 300 – 57 = **35.3**.

The slides' conclusion
Baker's FCF of $35,300 is positive but smaller than its $70,000 interest expense. Operations and investment are covered, but the firm is **not yet generating enough cash** to give creditors and shareholders an adequate return.

Sources/uses rule (Table 4.3): a decrease in assets, an increase in liabilities, profit, depreciation and selling stock = *sources*. An increase in assets, a decrease in liabilities, losses, dividends and repurchasing stock = *uses*. The change in retained earnings is not entered directly because it is already reflected through "net profit" and "dividends paid".

9 Cash flow within the month and the personal budget

Cash within the month. A cash budget only shows the position at month-end. The slides warn that this may not be enough: if wages are paid on the 1st but

customers only pay on the 25th, the firm can run short of cash mid-month even if month-end shows a surplus. So the firm needs to look at the *daily* pattern of receipts and payments and arrange a credit line before it is needed.

Personal budget (Personal Finance Example 4.10). The same logic applies to personal finance:

	Jan	Feb	Mar
Total income	$4,775	$4,775	$4,865
Total expenses	$4,026	$5,291	$7,396
Surplus/(deficit)	$749	−$516	−$2,531
Cumulative	$749	$233	**−$2,298**

The cumulative deficit of $2,298 must be covered by raising income, cutting expenses, drawing on savings or borrowing. Borrowing on a credit card can cover a deficit briefly but is dangerous if repeated. The first step of personal planning is setting clear goals, each with a priority, time frame and cost estimate (slide example: a master's degree costing $60,000 in 2 years, a condominium costing $250,000 in 6 years).

Key takeaways

1. Depreciation is a noncash charge; faster depreciation = lower early taxes.
2. Asset increase = cash out; liability increase = cash in.
3. OCF = EBIT(1–T) + Dep; FCF = OCF – NFAI – NCAI.
4. Cash budgets reveal borrowing needs in advance — arrange credit lines early.
5. Use scenarios — worst-case needs can far exceed the base case.
6. Percent-of-sales ignores fixed costs; split fixed vs variable.
7. External financing required is the balance-sheet plug.

Case studies

Case & worked answers

Group assignment: Indofood 5-year forecast (2026–2030)

Task: fill Indofood's assumptions (growth, operating cost ratios, non-operating items, tax & attribution, cash flow) and justify each versus the 2025 reference.

Links: sales growth is the key pro forma input; cost ratios = percent-of-sales (but not all costs are variable — wheat prices and the rupiah drive COGS; CPO affects agribusiness); working capital and capex = NCAI/NFAI logic; dividends must be cash-supported.

Pitfalls: optimistic growth with falling cost ratios unexplained; unsupported payouts; inconsistent capex and depreciation. Finally check margins, net income, cash and that the balance sheet balances.

Worked examples

Problem 1 · Medium · **OCF dan free cash flow**

EBIT Rp500 juta, pajak 25%, penyusutan Rp80 juta, investasi aset tetap bersih (NFAI) Rp150 juta, kenaikan modal kerja bersih (NCAI) Rp40 juta. Hitung OCF dan FCF.

1. NOPAT = EBIT × (1 – T) = 500 × 0,75 = 375.
2. `OCF = NOPAT + Penyusutan = 375 + 80 = 455`
3. `FCF = OCF - NFAI - NCAI = 455 - 150 - 40 = 265`

Answer: OCF Rp455 juta; FCF Rp265 juta (kas yang bebas dibagikan ke pemberi dana).

Problem 2 Medium **Garis lurus vs MACRS**

Mesin Rp100 juta, umur 5 tahun, tanpa nilai sisa. Tarif MACRS 5 tahun: 20%, 32%, 19,2%, 11,52%, 11,52%, 5,76%. Pajak 25%. Berapa penyusutan tahun 1 dan 2 dengan kedua metode, dan berapa pajak yang "tertunda" di tahun 2 bila memakai MACRS?

1. Garis lurus: 100 ÷ 5 = 20 per tahun → tahun 1 = 20, tahun 2 = 20.
2. MACRS: tahun 1 = 20% × 100 = 20; tahun 2 = 32% × 100 = 32.
3. Selisih penyusutan tahun 2 = 32 – 20 = 12.
4. Penghematan pajak = 12 × 25% = **Rp3 juta** di tahun 2. Total penyusutan tetap 100 — yang berubah hanya *waktunya*.

Answer: Tahun 2: MACRS 32 vs garis lurus 20 → pajak lebih rendah Rp3 juta (kas datang lebih awal).

Problem 3 Medium **Anggaran kas**

Penjualan Jan 100, Feb 120, Mar 150 (Rp juta). 30% tunai, 70% diterima bulan berikutnya. Pengeluaran Maret 140. Saldo kas awal Maret 20, saldo minimum 15. Perlu pinjam berapa?

1. Penerimaan Maret = 30% × penjualan Maret + 70% × penjualan Februari = 0,3 × 150 + 0,7 × 120 = 45 + 84 = 129.
2. Saldo akhir sebelum pinjaman = 20 + 129 – 140 = 9.
3. Kekurangan dari minimum = 15 – 9 = **6**.

Answer: Perlu pinjaman jangka pendek Rp6 juta di bulan Maret.

Problem 4 Easy **Depreciation tax shield**

This year's depreciation is IDR 80 million, tax rate 22%. How much tax does depreciation save? Why is faster depreciation worth more?

1. `Tax shield = Depreciation × Tax rate = 80 × 22% = IDR 17.6 million`
2. Depreciation uses no cash but lowers taxable income → tax paid falls by IDR 17.6 million.
3. Total depreciation over the asset's life is the same, but faster depreciation (MACRS) moves the tax savings into early years — and earlier money is worth more (time value of money).

Answer: Tax saving IDR 17.6 million.

Problem 5 Medium **Pro forma income: percent of sales vs fixed costs**

This year's sales are IDR 200 million: COGS 150 (fixed 60 + variable 45% of sales), operating expenses 20 (fixed 10 + variable 5%), interest 4 (fixed). Next year's sales are forecast at IDR 240 million. Compute next year's profit before tax with (a) the percent-of-sales method, (b) a fixed/variable split.

1. (a) This year's ratios: COGS 150/200 = 75%, operating expenses 10%, interest 2%. Profit before tax = 240 – 180 – 24 – 4.8 = **31.2**.
2. (b) COGS = 60 + 45% × 240 = 168. Operating expenses = 10 + 5% × 240 = 22. Interest = 4.
3. Profit before tax = 240 – 168 – 22 – 4 = **46**.
4. A gap of 14.8: percent of sales assumes every cost rises 20%, but fixed costs do not → profit is *understated* (as in the Fair Traders case on the slides).

Answer: (a) IDR 31.2 million; (b) IDR 46 million. Method (b) is more accurate.

Problem 6 Medium **Pro forma balance sheet: external financing (plug)**

Forecast for next year-end (IDR million): total assets 520. Forecast liabilities (payables, taxes, existing bank debt) 210. Opening equity 250, extra retained earnings from the pro forma income statement 30. How much external financing is needed?

1. Ending equity = 250 + 30 = 280.
2. Funding already in place = 210 + 280 = 490.
3.
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External financing = Total assets - (Liabilities + Equity)
= 520 - 490 = 30
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In the judgmental approach this is the "plug" that balances the sheet.
4. A negative result means surplus funds that can repay debt or be invested.

Answer: IDR 30 million of external financing is needed.

Practice questions

Try answering first. The answer key is below each question.

1. Depreciation is a noncash charge because…

A. It doesn't reduce profit
B. It reduces profit but no cash leaves that year
C. Paid in cash yearly
D. Balance sheet only

Answer: B. Cash left when the asset was bought.

2. Coulson: November ending cash −51, minimum 25. Required financing…

A. 25
B. 51
C. 76
D. 98

Answer: C. From −51 to 25 = 76.

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3. OCF 392.3, NFAI 300, NCAI 57. FCF…

A. 35.3
B. 92.3
C. 449.3
D. −35.3

Answer: A. 35.3.

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4. Why can percent-of-sales understate profit when sales grow?

A. Ignores tax
B. Treats all costs as variable
C. Uses old prices
D. Profit can't be projected

Answer: B. Operating leverage.

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5. In a judgmental pro forma balance sheet, external financing required is…

A. Retained earnings
B. The balancing plug
C. Fixed assets
D. Revenue

Answer: B. Closes the gap.

Sources

- Materi Week 3 - MK02.pdf (Zutter & Smart, Principles of Managerial Finance 15e, Bab 4)
- Pre-Assignment Week 3 Instructions - Indofood Forecast
- Indofood_5Year_Forecast_Simple_Worksheet.xlsx